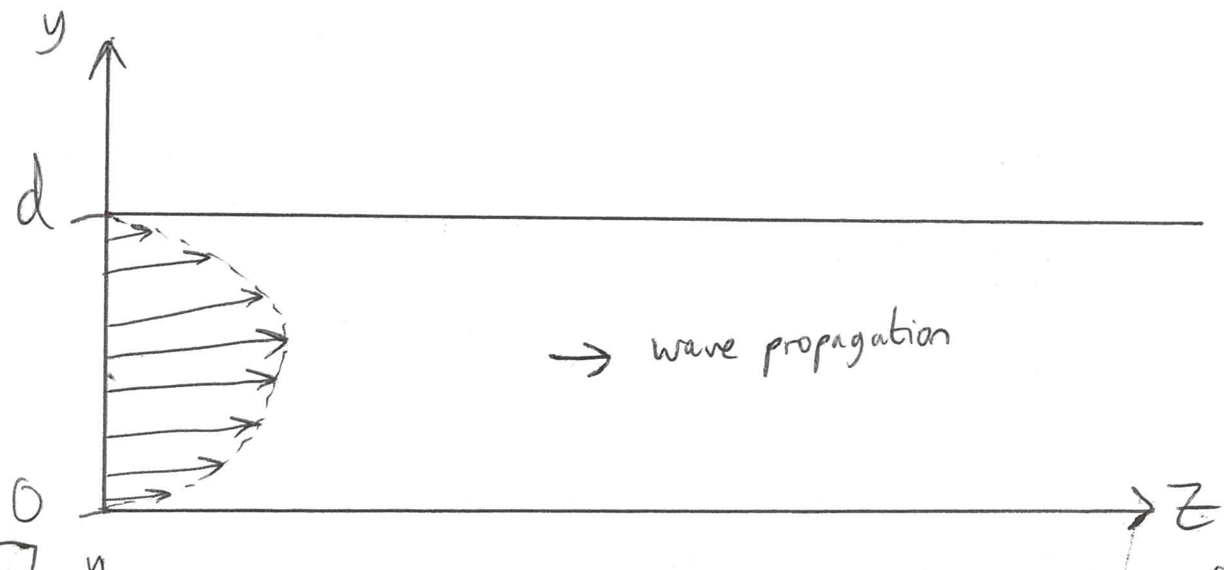
$$E_z(x,y) = A\sin(k_c y) \qquad k_c = \frac{n\pi}{d} \qquad n = 0, 1, 2, \ldots$$

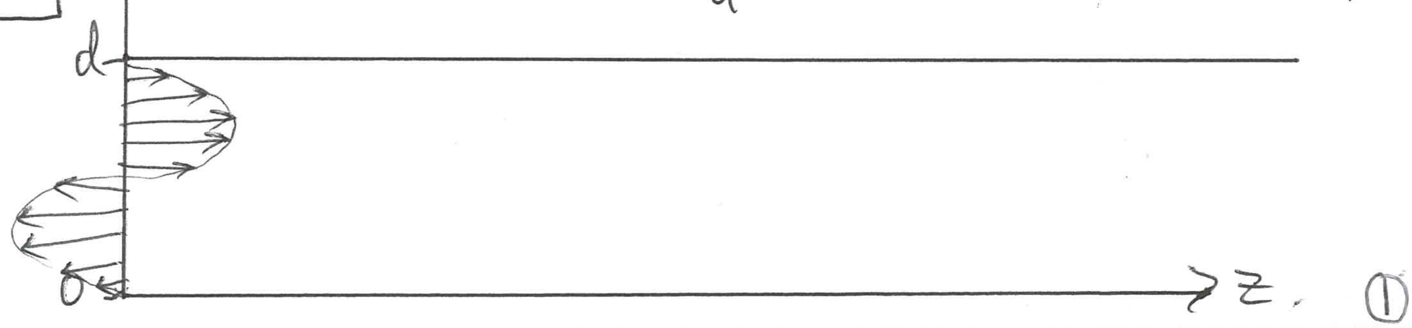
e.g. $n=1$ $\quad E_z(x,y) = A\sin\left(\frac{\pi y}{d}\right) \Big\} = 0$ at $y=0$, $y=d$

Side view: E_z:

$\boxed{TM_1}$



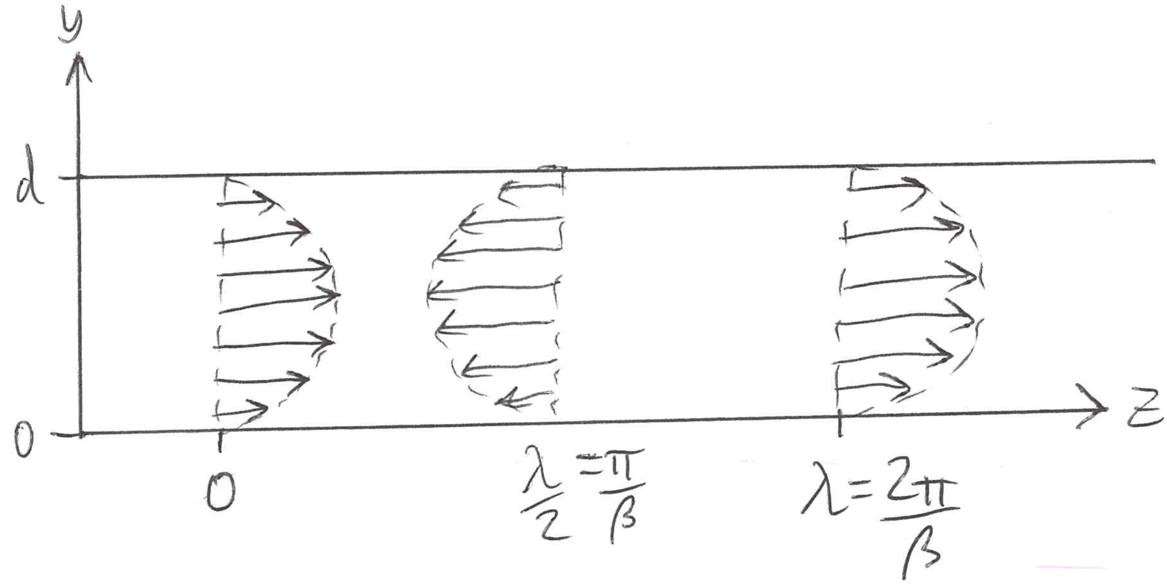
$\boxed{TM_2}$ $\quad n=2 \quad k_{c2} = \frac{2\pi}{d} \quad E_z(x,y) = A\sin\left(\frac{2\pi y}{d}\right)$



$$E_z(x,y,z) = E_z(x,y)\, e^{-j\beta z}$$

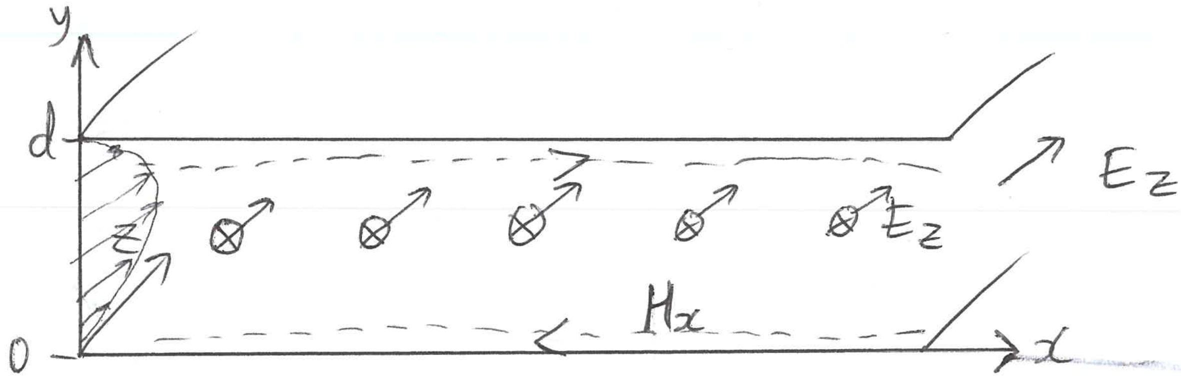
Side view:

$\boxed{TM_1}$



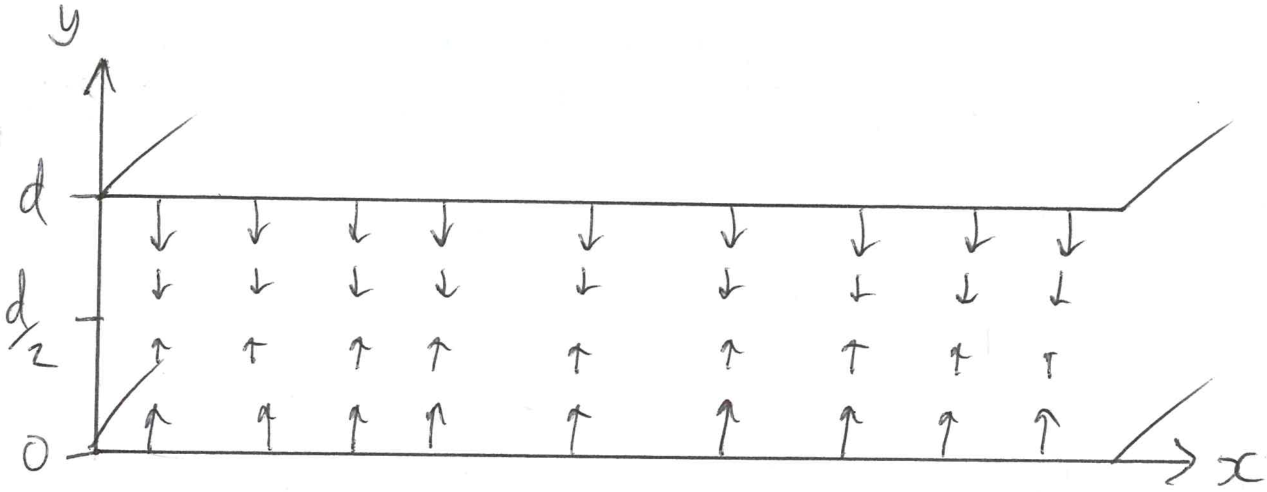
$$H_z = 0$$

Front view:

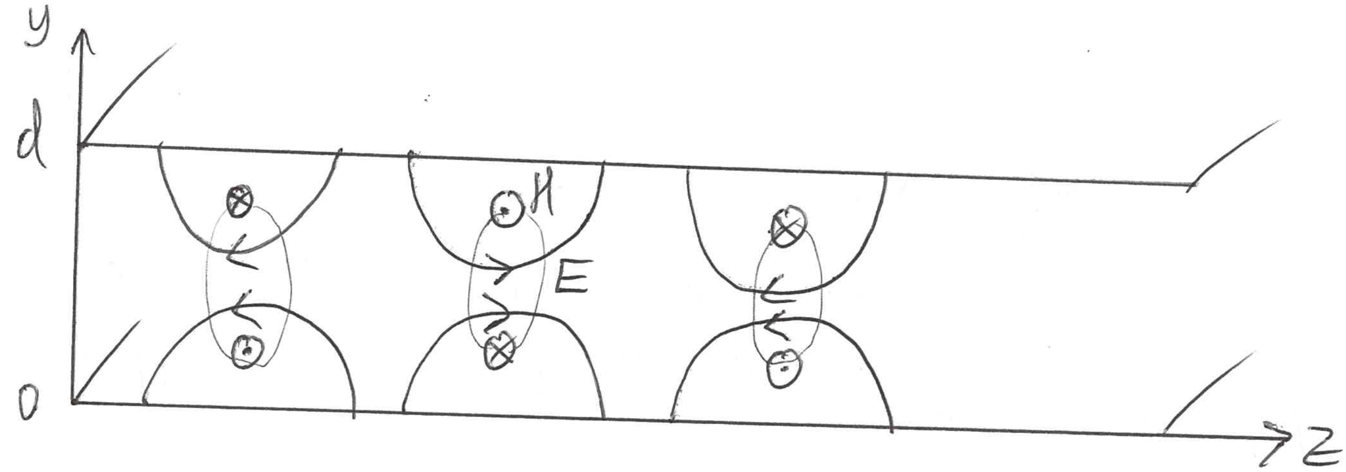


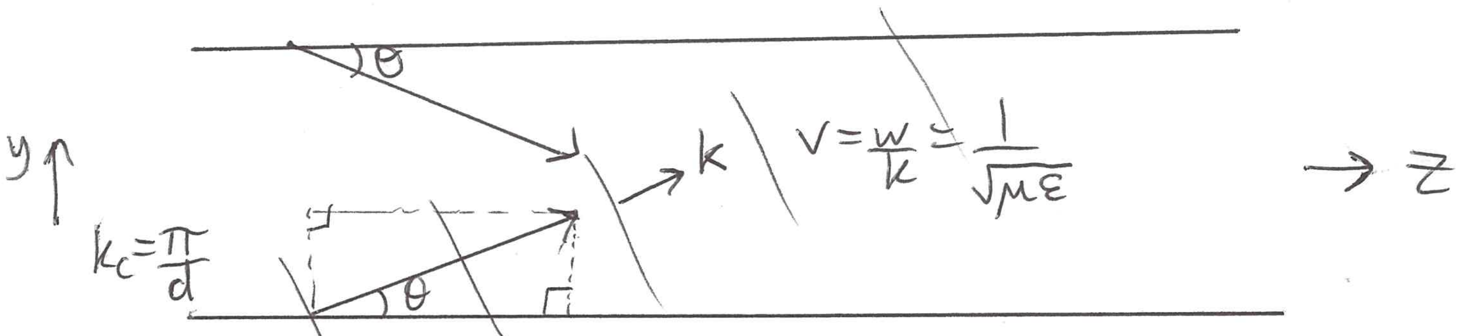
$$H_x = \frac{j\omega\varepsilon}{k_c} A_n \cos\left(\frac{n\pi y}{d}\right) e^{-j\beta_n z}$$

$$E_y = \frac{-j\beta_n}{k_c} A_n \cos\left(\frac{n\pi y}{d}\right) e^{-j\beta_n z}$$



Side view:





$k_c = k\sin\theta \qquad \beta_1 = k\cos\theta$

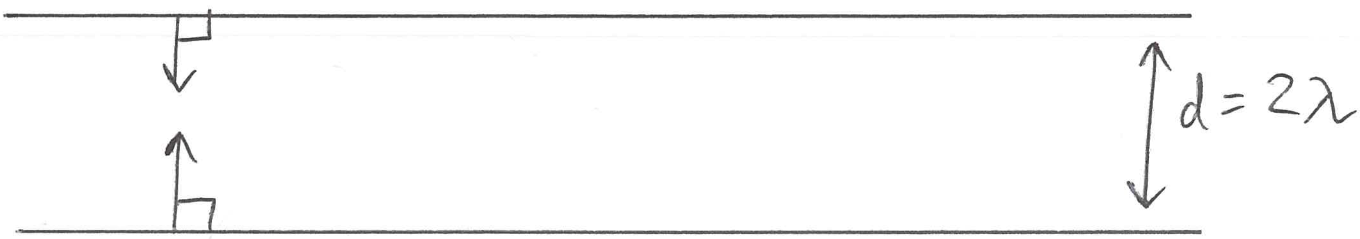
$$V_z = \frac{w}{\beta_1} = \frac{w}{k\cos\theta} = \frac{v}{\cos\theta} = \frac{1}{\sqrt{\mu\varepsilon}\cos\theta}$$

$0 \le \theta \le \pi$

$$v \le V_z \le \infty$$

At cutoff frequency: $k = k_c \Rightarrow w = w_c$

$\beta_1 = \sqrt{k^2 - k_c^2} = 0 \qquad \cos\theta = 0 \Rightarrow \theta = 90^\circ$



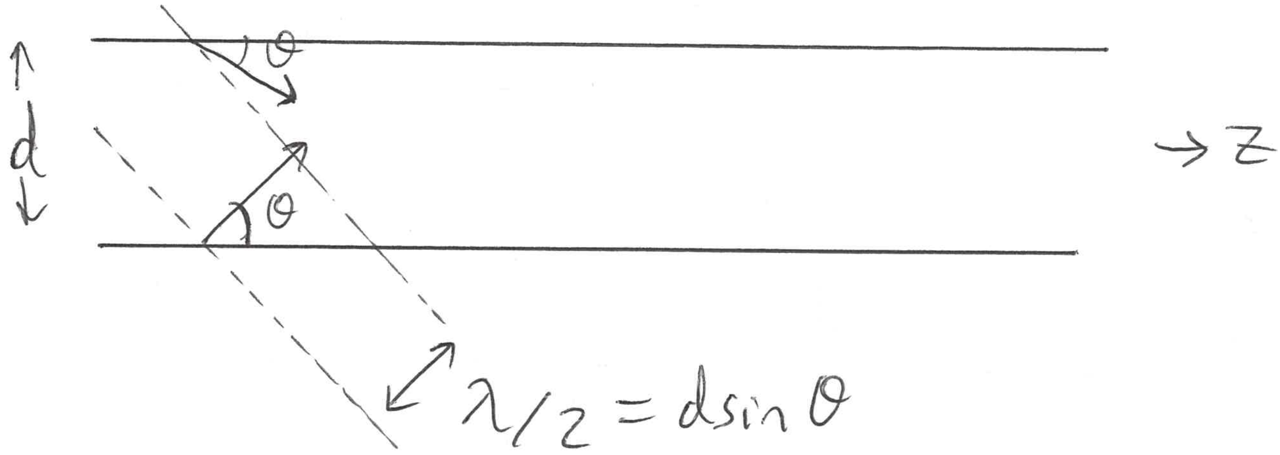
$k = \frac{2\pi}{\lambda} \qquad k_c = \frac{\pi}{d}$

$\Downarrow$

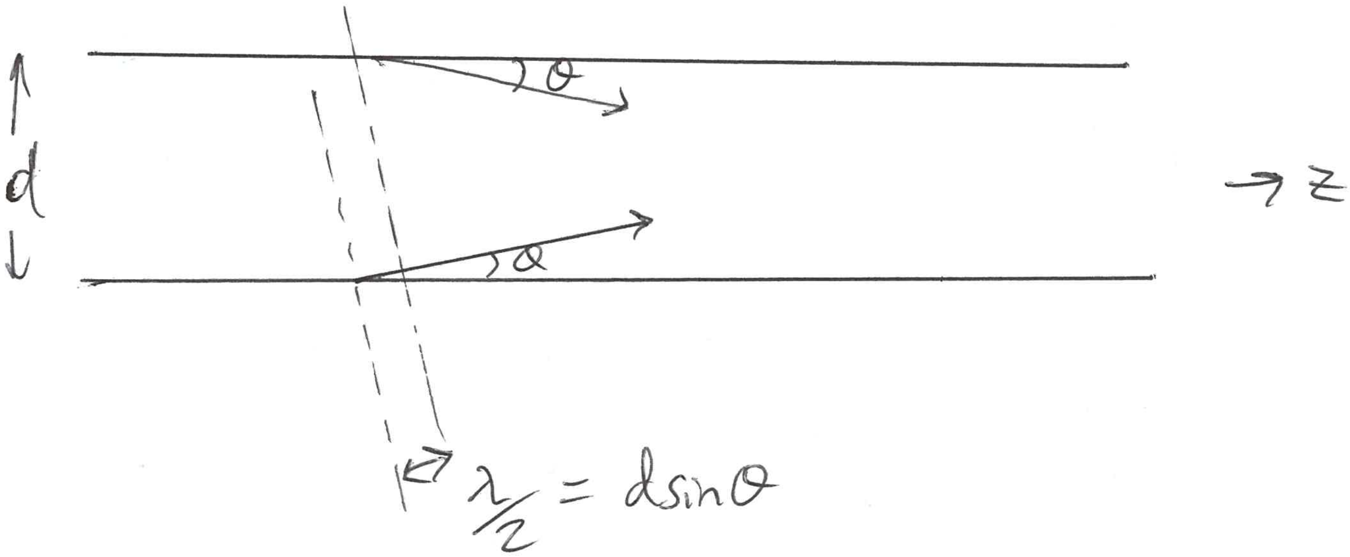
$\frac{2\pi}{\lambda} = \frac{\pi}{d} \Rightarrow \frac{\lambda}{2} = d$

$$\frac{1}{\sqrt{\mu\varepsilon}} = v = f\lambda = \frac{\omega}{2\pi}\lambda$$

$\omega > \omega_c$



$\omega \gg \omega_c$



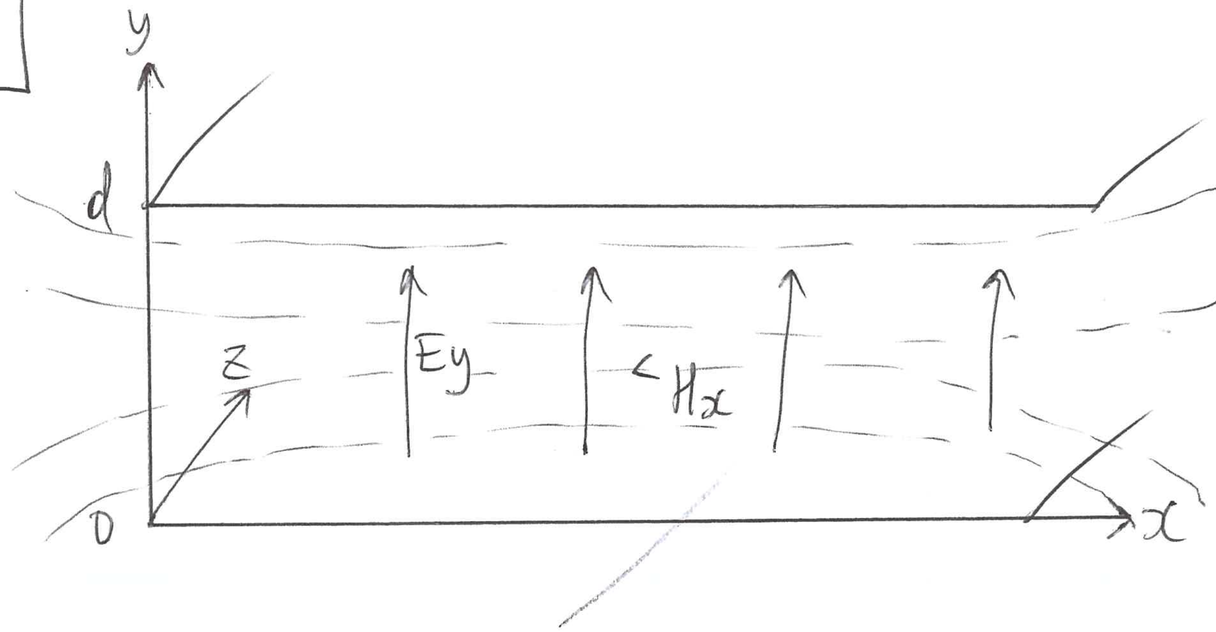
TM modes: $$E_z(x,y,z) = A_n \sin\left(\frac{n\pi y}{d}\right) e^{-j\beta_n z}$$

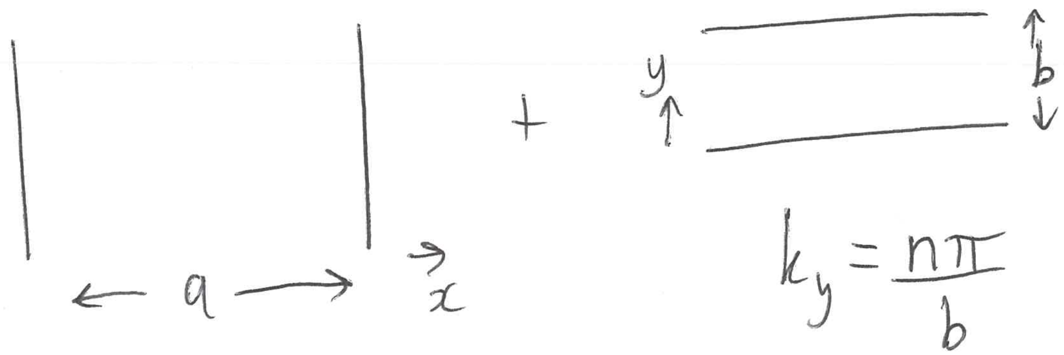
$n=0$: $$E_z(x,y,z) = 0$$

$$\Downarrow$$

$$TM_0 \equiv TEM$$

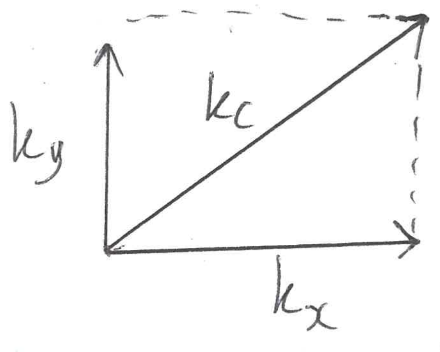
$\boxed{TM_0}$





$$k_x = \frac{m\pi}{a}$$

$$k_y = \frac{n\pi}{b}$$

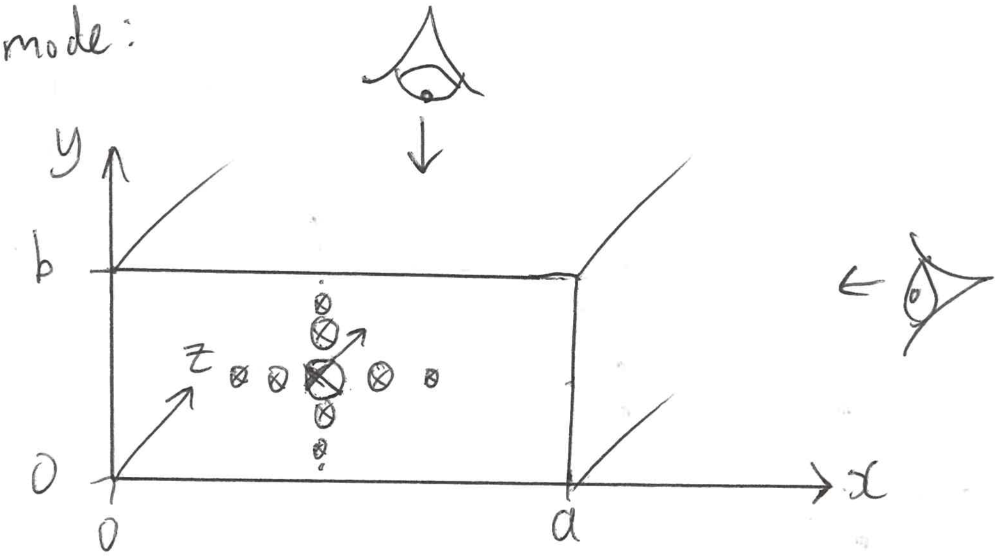


$$k_x^2 + k_y^2 = k_c^2$$

mn modes

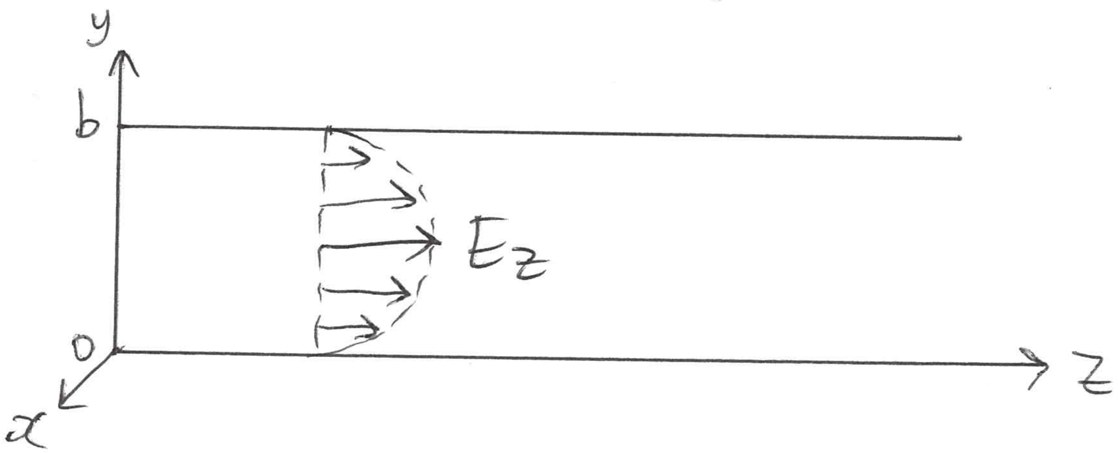
TM mode:

$\boxed{TM_{11}}$

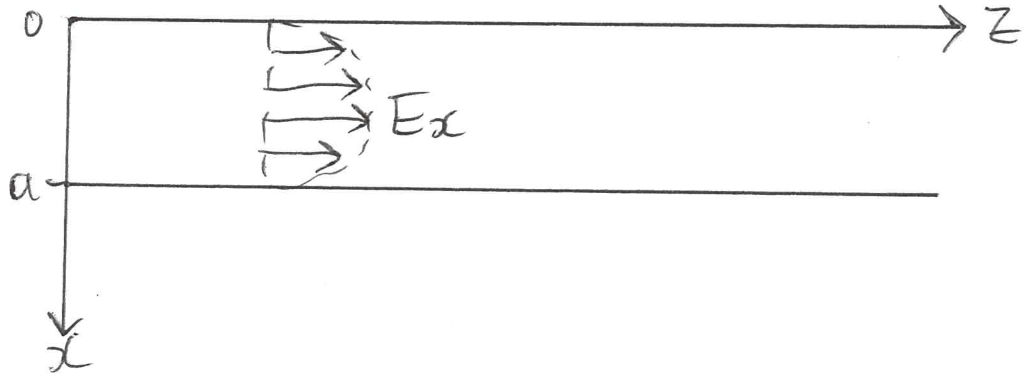


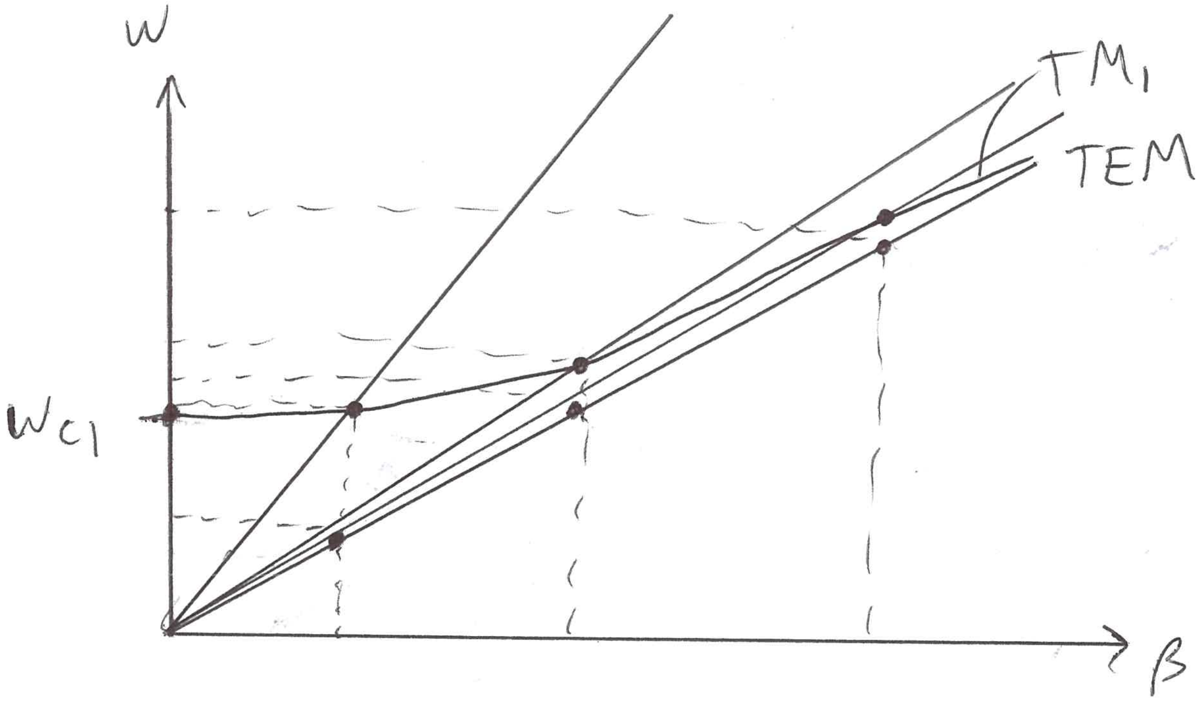
$$TM_{mn} \begin{cases} m \ \frac{1}{2}\lambda_s \text{ in } x \text{ direction} \\ n \ \frac{1}{2}\lambda_s \text{ in } y \text{ direction} \end{cases}$$

$\boxed{TM_{11}}$ side view:



Top view:





Phase velocity $\quad V_p = \dfrac{w}{\beta(w)}$

$w \to \infty$

$V_p \to V = \dfrac{1}{\sqrt{\mu\varepsilon}}$

$w = w_{c1}$

$V_p = \infty$